

Linux System Administration

Logging & System Documentation Lab Report

Date:	April 17, 2026
Topic:	System Logging, Documentation & Remote Access (SSH)
Platform:	Ubuntu / Linux

Executive Summary

This laboratory exercise focused on mastering Linux system documentation, SSH remote access, and troubleshooting tools. Through 10 hands-on tasks, critical system administration skills were developed including manual page navigation, system configuration discovery, and remote host management. All objectives were successfully completed, demonstrating proficiency in essential Linux sysadmin workflows.

Learning Objectives Achieved

- 1. Navigate and search Linux manual pages (man command) and apropos database
- 2. Identify system administration commands and their manual sections
- 3. Configure system hostname using hostnamectl
- 4. Troubleshoot SSH connectivity with verbose output
- 5. Manage manual page indexing and database maintenance
- 6. Perform remote SSH authentication and file operations

Key Competencies Demonstrated

Manual Pages & Documentation	Mastery of man command, section identification, grep filtering
System Configuration	hostnamectl usage, hostname management
Remote Access (SSH)	Authentication, verbose debugging, key fingerprint validation
File Management	Remote file creation, hidden file identification, permissions
Database Maintenance	mandb indexing, apropos troubleshooting

Practical Applications

These skills are fundamental for Linux system administrators: troubleshooting infrastructure issues, managing remote servers, accessing system documentation, and maintaining production environments.

Task Completion Details

Task	Objective	Result	Status
1	Find SSH version option	Identified -V flag via man pages	✓
2	Change system hostname	hostnamectl command identified	✓
3	Refresh manual page index	mandb command executed	✓
4	SSH debugging option	ssh -v option discovered	✓
5	Shell autocompletion	Tab key completion verified	✓
6	Manual section identification	Section 8 (System admin) found	✓
7	Hidden file enumeration	Identified 2 hidden files (.myfile1, .myfile2)	✓
8	Remote file creation via SSH	Successfully created /home/bob/myfile on node01✓	
9	Apropos database repair	Troubleshooted mandb indexing issue	✓
10	Configuration file discovery	Located nfsmount.conf (5) - NFS configuration	✓

Core Commands & Tools Mastered

Command-line Tools: man, apropos, mandb, hostnamectl, ssh, ssh -v, touch, ls -la, grep, sudo

Technical Skills Gained

System Administration: Manual page navigation, system configuration discovery, hostname management

Remote Access & Security: SSH authentication, key fingerprint validation, remote command execution

Troubleshooting: Verbose output analysis, database indexing issues, apropos search problems

File System: Permission analysis, hidden file identification, remote file creation

Documentation: Effective use of manual pages as primary reference material

Path Forward

Completion of this laboratory establishes a strong foundation for advanced Linux system administration studies, including kernel management, service configuration, network administration, and system performance tuning. These fundamentals will support progression toward RHCE, LPIC, and professional sysadmin certifications.